

Quantifying Modality Inertia in Medical Vision-Language Models: Image Contribution Metric, Gate Stagnation, and the Failure of Debiasing Under Label Circularity

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ABSTRACT

Objective: Medical vision-language models may obtain high scores from radiology reports without using corresponding images. This risk is acute when evaluation labels and model text inputs share report provenance. We evaluated whether text-debiasing increased incremental image information in a MIMIC-CXR-derived classification task and assessed an Image Contribution Metric (ICM) as a modality-use audit.

Methods: We analysed 2,953 frontal studies assigned at patient level to training (2,049), validation (449), and test (455) sets. Five fusion conditions were evaluated across eight seeds: baseline gated fusion, a learned text-prior product-of-experts variant, an image-noise variant, an exploratory grounding-aware variant, and an adaptive-gate variant. The primary intervention replaced the radiograph with a constant gray image while preserving the report. ICM measured the relative Macro-F1 difference between the full model and this Gray-Square control. Seed-level inference included signed-rank tests, two one-sided tests within an operational equivalence interval of -0.01 to 0.01, bootstrap intervals, calibration analysis, and retained checkpoint-gate inspection.

Results: Baseline Macro-F1 was 0.9397 ± 0.0061 ; the retained Gray-Square control achieved 0.9381. Baseline ICM was 0.0016 ± 0.0065 (95% bootstrap confidence interval $[-0.0026, 0.0061]$). The signed-rank test was non-significant ($p = 0.4219$), whereas the equivalence test supported an ICM inside the operational interval ($p = 0.0040$). No intervention produced a robust ICM increase. The text-prior variant increased expected calibration error from 0.0237 to 0.0621 and Brier score from 0.0522 to 0.0685. Retained baseline gates remained near their class-specific initial values, with an overall mean of 0.49975.

Conclusions: In this report-linked benchmark, near-ceiling classification was compatible with negligible incremental image contribution. ICM and gate inspection expose a construct-validity problem that aggregate performance alone cannot detect. The result is specific to this cohort and archived implementation; independent labels, matched controls, corrected preprocessing and objectives, provenance locking, and external validation are required before clinical or general claims.

Statement of significance

Biomedical informatics studies increasingly combine images and text, but a multimodal score does not reveal which input supplied the decisive information. This study addresses a specific threat to validity: CheXpert targets and the model's report input share linguistic provenance. We combine a report-preserving image intervention, a seed-level ICM, calibration analysis, and retained-gate inspection to distinguish multimodal performance from incremental visual evidence. The audit shows how apparently strong performance can persist after removing image information and provides a reproducible reporting pattern for benchmark designers, clinical-AI evaluators, and researchers assessing modality reliance. ICM is proposed as a diagnostic measure, not a standalone certificate of grounding.

1. Introduction

Medical vision-language models (VLMs) offer a natural way to combine the complementary information in radiographs and radiology reports. Large paired resources such as MIMIC-CXR support representation learning, classification, report generation, and clinical question answering at a scale that would be difficult to obtain through manual annotation alone [1]. Contrastive and local-alignment approaches, including ConVIRT, GLoRIA, and BioViL, have shown that report semantics can improve biomedical image representations [2–4]. The clinical interpretation of

a downstream score nevertheless depends on a more basic question: did the declared image input materially affect the model’s decision?

High accuracy cannot answer that question. Deep networks can learn predictive features that are stable within a benchmark but unrelated to the intended mechanism [5]. In radiology, acquisition hardware, population composition, annotation practice, and clinical context can each create shortcuts [6]. Multimodal systems introduce an additional route: a strong language pathway may dominate a weaker visual pathway. Correct answers can then coexist with flawed image rationales, stable outputs can survive removal of the image, and a plausible heatmap can be mistaken for evidence of causal grounding [7–9]. Recent chest-radiography studies further show that performance claims can change when the amount or relevance of clinical context changes [10]. These observations make modality contribution an empirical estimand rather than an architectural assumption.

Label provenance makes the problem sharper in MIMIC-CXR-derived tasks. CheXpert maps radiology-report language to uncertainty-aware pathology labels [11]. If the same report, or its findings field, is supplied to the predictor, the text input and target share an information source. A model can therefore be rewarded for recognizing language that helped produce the label. Report-conditioned prediction may still serve a useful informatics purpose, but it is not equivalent to independent image-based recognition. Natural-language labelers also have their own error modes and potential subgroup effects, which can propagate into any downstream evaluation [12, 13].

Standard debiasing methods do not automatically resolve this circularity. Visual-question-answering methods reduce language or answer priors by penalizing an easy unimodal pathway or combining it with a multimodal expert [14–16]. Their logic assumes that the bias pathway is a nuisance proxy rather than a near-direct encoding of the benchmark target. When report text is also the label source, suppressing the text pathway may discard information rewarded by the endpoint without creating an independent visual signal. A lower score then shows that text mattered, not that image use improved. Prototype and counterfactual methods can produce more interpretable predictions, but explanation quality remains distinct from behaviour under a controlled input intervention [17].

Three evidentiary gaps remain. First, few studies replace the radiograph while holding report, label, architecture, and test cohort fixed. Second, a balanced fusion gate or attention coefficient is often treated as attribution even though it may remain near initialization [18]. Third, non-significance is sometimes described as proof of no modality contribution, although failure to reject zero and evidence of equivalence are different statistical conclusions [19]. Together, these gaps can make a nominally multimodal model appear visually grounded when the evaluation has not tested that claim.

We address these gaps in a controlled audit of a three-class MIMIC-CXR-derived task comprising Normal, Pneumonia, and Pneumothorax. We compare the full model with a report-preserving Gray-Square intervention, text-only and image-only controls, and four fusion or text-debiasing variants alongside the baseline gated model. The primary diagnostic is the Image Contribution Metric (ICM), the relative Macro-F1 difference between a full VLM and its Gray-Square reference. We pair ICM with an operational equivalence test, calibration metrics, class-specific performance, gate inspection, and qualitative Grad-CAM [20]. The study asks whether removing image information changes predictive performance, whether the tested interventions increase incremental image contribution, and whether retained gates provide evidence of learned visual reliance. It is a benchmark audit, not a new clinical predictor and not a claim that radiographs are generally unnecessary.

2. Related work

Medical image-text learning and report-derived targets. Paired image-report learning has improved transfer and label efficiency by using free text as a rich source of supervision. ConVIRT uses contrastive learning across paired views; GLoRIA models global and local correspondences; and BioViL emphasizes biomedical text semantics [2–4]. Clinical BERT representations similarly improve the handling of biomedical terminology, negation, and context [21, 22]. These advances establish the utility of text but do not by themselves identify the incremental contribution of the image in a downstream task. That question is especially consequential when labels are automatically extracted from the same reports used during prediction.

Shortcut learning and unimodal bias. Shortcut learning describes reliance on an easy predictive signal that performs well in-distribution but does not match the intended construct [5]. Language priors are a long-standing example in visual question answering. VQA-CP, ensemble-based debiasing, product-of-experts objectives, and RUBi attempt to reduce this dependence by exposing or down-weighting the language-only expert [14–16]. In a medical benchmark with report-derived targets, however, the language pathway is not merely correlated with the outcome:

it participates in the outcome-generation process. The present study evaluates whether methods borrowed from the former setting retain their intended meaning in the latter.

Auditing modality reliance. Recent evaluations separate output correctness from evidence dependence. Multimodal answers can have unreliable visual rationales, and predictions can remain consistent even when the image is uninformative [7, 9]. Controlled image removal, cross-patient replacement, and text-only comparisons have shown that several chest-radiography VLMs retain substantial performance without the image [8]. Clinical context can also alter apparent model quality [10]. These studies motivate behavioural interventions but generally do not combine a per-model relative contribution metric, equivalence testing, seed-level calibration, and gate inspection in one audit.

Gates and explanation limits. Gated multimodal units provide a compact mechanism for combining modality representations [18]. A gate value is nevertheless a model parameter, not a causal estimate. A non-zero gate may be unidentifiable when both pathways encode redundant signals, or it may stay near initialization if the objective supplies little incentive to move it. The same caution applies to attention and saliency: a visually plausible map does not show that removal of the highlighted evidence changed the output. Reasoning-oriented radiography models and prototype methods may improve transparency, but their explanations remain complementary to input interventions [17, 23]. We therefore position ICM as an intervention-based diagnostic and treat gates and Grad-CAM only as supporting observations.

3. Methods

3.1. Study design, data source, and cohort

We performed a retrospective computational audit. The estimand was incremental image contribution when the report remained available. The principal intervention replaced each radiograph by a constant gray tensor while retaining the report, target, fusion path, and evaluation cohort. Full multimodal outputs were compared with this report-preserving control and with descriptive text-only and image-only controls. Figure 1 connects cohort provenance, the model pathways, controlled ablations, and seed-level inference.

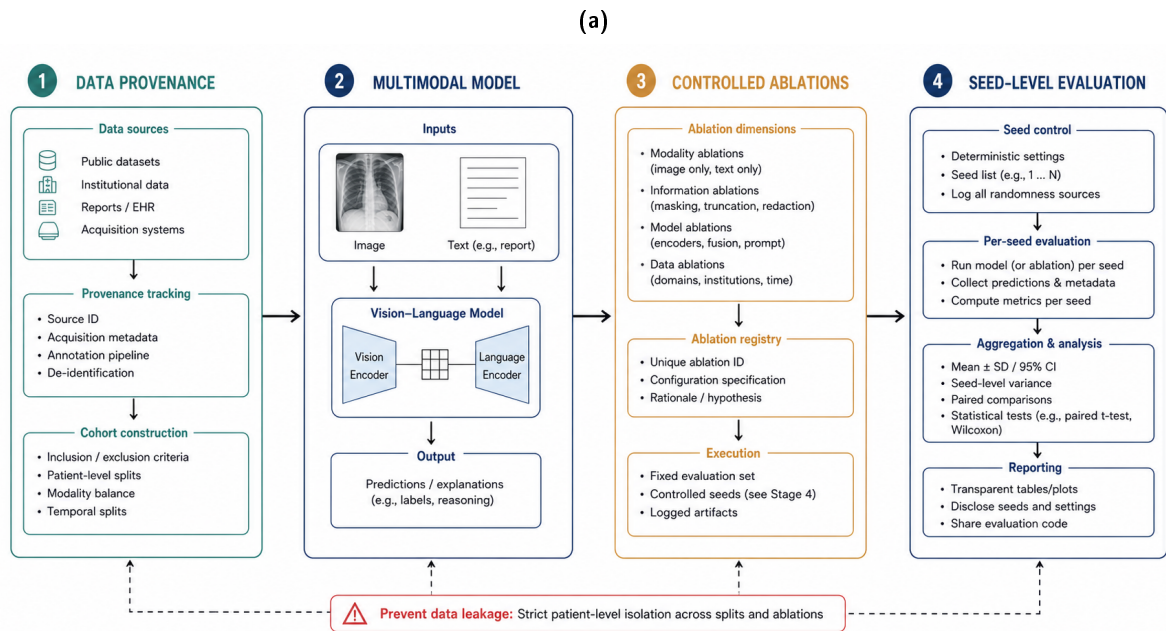
MIMIC-CXR JPG images and reports were accessed through official PhysioNet credentialing under the applicable Data Use Agreement [1]. The source flow comprised 377,110 images in 227,835 studies; 227,827 studies had CheXpert labels and 108,626 met the frontal-view eligibility criteria. An analysis cap selected up to 2,500 Normal, 2,500 Pneumonia, and 1,000 Pneumothorax candidates. Of 2,962 attempted downloads, nine JPG files were unavailable, leaving 2,953 usable studies. The inherited patient-level GroupShuffleSplit assigned 2,049 studies to training, 449 to validation, and 455 to test, with no subject overlap. The test set contained 240 Normal, 168 Pneumonia, and 47 Pneumothorax studies.

CheXpert generated the targets from report language using the project’s uncertain-to-negative mapping [11]; the model received the same report’s findings field. The experiment consequently audits label-report circularity rather than an independent radiograph-only diagnostic task. The replication unit for seed-level inference was a trained model at a fixed seed. Baseline CADQ, LearnedMixinFusion (LMH), Noise-Consistency, GACR-B, and Adaptive-CADQ each had eight retained seeds: 42, 123, 456, 789, 1010, 1111, 1212, and 1313. The eight-seed plan covered baseline, LMH, and Noise-Consistency; Adaptive-CADQ was extended from three to eight seeds after the initial run, and GACR-B was added post hoc. The latter two analyses are therefore exploratory.

3.2. Preprocessing and architecture

The findings field was tokenized with emilyalsentzer/Bio_ClinicalBERT, truncated or padded to 256 tokens, and randomly replaced by an empty string in 10% of training examples. Test and validation text was not masked. Images were converted to one channel, resized to 256×256 , randomly cropped to 224×224 during training, and mapped from $[0, 255]$ to $[-1, 1]$. Training augmentation used horizontal flipping, rotation up to 10° , translation, and brightness/contrast jitter; the minority-class transform increased the rotation, translation, scale, shear, jitter, and blur ranges. Validation and test preprocessing was deterministic. The archived $[-1, 1]$ mapping may not match the native TorchXRyVision normalization expected by the selected weights; this is an implementation limitation requiring a corrected rerun [24].

The image branch used a TorchXRyVision DenseNet-121 with densenet121-res224-mimic_ch weights [24, 25]. Its convolutional stem and first dense block were frozen, and its output was projected to 512 dimensions through a linear layer, LayerNorm, Gaussian Error Linear Unit, and dropout (0.25). The text branch used Bio_ClinicalBERT with the first four transformer layers frozen and an analogous 512-dimensional projection [21, 22]. One projected



(b)

Cohort construction and patient-level analysis split

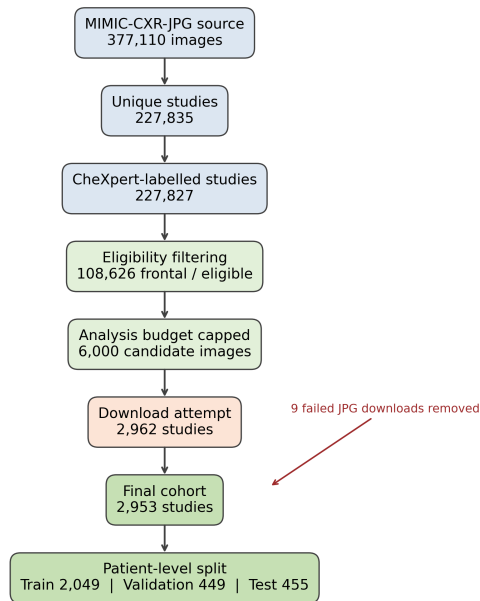


Figure 1: Study design. (a) Methodological framework linking data provenance, paired image-report inputs, controlled ablations, and evaluation. The audit compares five fusion conditions over eight seeds and separates predictive performance from incremental image contribution. (b) Cohort-construction flow. Nine unavailable JPG files were removed before the final 2,953-study cohort was partitioned at patient level.

image token was used as the query and one projected text token as the key/value in an eight-head attention block with residual LayerNorm. Because there is only one key, this operation cannot express word-region alignment; we interpret it as feature interaction, not causal attention.

Baseline CADQ combined image and text logits through a bounded learnable per-class gate [18]:

$$\alpha_c = 0.20 + 0.60 \text{ sigmoid}(u_c), \quad z_c = \alpha_c z_c^{\text{img}} + (1 - \alpha_c) z_c^{\text{text}}. \quad (1)$$

The initial logits $u = (-0.5, 0, 0.5)$ corresponded to image-gate values (0.426524, 0.500000, 0.573476) for Normal, Pneumonia, and Pneumothorax.

LMH added a five-fold out-of-fold term-frequency/inverse-document-frequency logistic text prior (5,000 uni-gram/bigram features, stop-word removal, balanced weights), temperature calibration, $\epsilon = 0.1$ smoothing, an input-dependent prior gate, and an entropy penalty of 0.05 annealed over three epochs. It was warm-started from CADQ and followed the product-of-experts rationale used for unimodal-bias reduction [16]. Noise-Consistency perturbed the image with Gaussian noise ($\sigma = 0.25$). Its archived objective added $-D_{KL}(p_{\text{clean}} \| p_{\text{noisy}})$ to a minimized loss and therefore rewards divergence; throughout this manuscript it is treated as an anti-consistency or noise-sensitivity condition, not a valid conventional consistency objective. Adaptive-CADQ replaced the static per-class gate with a sample-dependent scalar computed from image and text entropies by a $2 \rightarrow 16 \rightarrow 1$ multilayer perceptron. Exploratory GACR-B used an out-of-fold text baseline and a cosine penalty of 0.3 on examples for which that baseline was wrong, following the grounding-aware intervention of Sadanandan and Behzadan [9].

The image-only control used the image encoder with a 512-to-256 classifier; the text-only control used frozen Bio_ClinicalBERT with a 768-to-256 classifier. The Gray-Square control replaced every image by a constant intensity-128 tensor but preserved the report and multimodal architecture. Only one retained run was available for each control. Gray-Square is therefore a report-preserving image ablation, not an independently trained eight-seed text-only ensemble.

3.3. Training and outcomes

Models were trained for 10 epochs with AdamW and weight decay 10^{-4} [26]. Learning rates were 10^{-5} for the image and text backbones, 10^{-4} for fusion and LMH heads, and 5×10^{-5} for static gates. OneCycleLR used 10% warm-up [27]. Physical batch size was 8 with four gradient-accumulation steps (effective batch 32), mixed precision, and gradient scaling. The objective combined focal cross-entropy ($\gamma = 1.5$), label smoothing (0.05), and effective-number class weights ($\beta = 0.99$) [28]. Class-aware sampling replicated Pneumothorax examples up to fivefold and Pneumonia examples up to threefold. The highest validation Macro-F1 checkpoint was restored for multimodal conditions. The archived image-only pipeline did not consistently restore its best checkpoint and is reported as descriptive.

The primary diagnostic was

$$\text{ICM} = \frac{F1_{VLM} - F1_{Gray}}{F1_{VLM}}, \quad (2)$$

where positive values indicate better Macro-F1 with the real radiograph. The relative form was chosen to express the incremental change on the scale of the full model. An ICM near zero indicates that replacing the image did not materially change this endpoint; it does not prove that the image encoder is universally uninformative. The analysis used an operational equivalence interval of $[-0.01, 0.01]$. This project-defined margin is neither a validated clinical threshold nor a regulatory acceptance criterion.

Secondary outcomes were Macro-F1, accuracy, one-versus-rest Macro-AUROC, per-class F1 and sensitivity, ten-bin expected calibration error (ECE), multiclass Brier score, row-normalized confusion matrices, retained image-gate values, and qualitative Grad-CAM. ECE and Brier were recomputed from retained probability arrays when absent from the summary dictionary; missing values were never replaced with zero. Gate values were taken from the per-seed checkpoint export in `q1_audit_results.json`. The displayed gate figure uses all eight exported values and their equation-derived initialization references, rather than hard-coded aggregate values.

3.4. Statistical analysis and reproducibility

Seed-level differences used paired Wilcoxon signed-rank tests with Pratt zero handling and paired t -tests. ICM summaries included the median, standard deviation, Cohen’s d , and percentile bootstrap 95% confidence intervals

from 2,000 resamples. Two one-sided tests (TOST) evaluated whether ICM lay within $[-0.01, 0.01]$ [19]. Because the same 455 studies recur at every seed, test subjects were not concatenated across seeds for confirmatory inference.

DeLong tests compared correlated Pneumothorax receiver-operating-characteristic curves within the retained seed-42 probability set [29]. McNemar tests compared paired classification decisions on the same 455 seed-42 studies [30]. Benjamini-Hochberg correction controlled the false-discovery rate within each test family [31]. A McNemar p -value of 1 denotes balanced discordance, not identical predictions; the two off-diagonal counts are therefore reported with the test result.

Python, NumPy, and PyTorch seeds were fixed and deterministic cuDNN settings were enabled. Run metadata, validation histories, checkpoints, predictions, probabilities, and configurations were retained. The fixed merge archive resolved duplicate keys by the first-seen value, but the earlier workflow did not preserve a deterministic source-priority rule and content-hash manifest. Results are consequently tied to this archive snapshot. Reporting was audited against CLAIM and TRIPOD+AI principles [32, 33]; checklist summaries and a provenance manifest are included in the Supplementary material.

4. Results

4.1. Analysis set, discrimination, and calibration

The fixed archive contained 40 condition-seed records: five conditions with eight seeds each, evaluated on the same 455 held-out studies. Baseline CADQ achieved Macro-F1 0.9397 ± 0.0061 , accuracy 0.9607 ± 0.0032 , and Macro-AUROC 0.9974 ± 0.0004 (Table 1). LMH had the highest mean Macro-F1 (0.9435 ± 0.0093) but lower and more variable AUROC (0.9849 ± 0.0113). Noise-Consistency had the lowest Macro-F1 (0.9346 ± 0.0075). Exploratory GACR-B reached 0.9417 ± 0.0059 , and Adaptive-CADQ remained close to baseline at 0.9388 ± 0.0072 .

Table 1

Performance and calibration across eight seeds (mean $\pm$ standard deviation). GACR-B was added post hoc and is exploratory.

Condition	Macro-F1	Accuracy	Macro-AUROC	ECE	Brier
Baseline CADQ	0.9397 ± 0.0061	0.9607 ± 0.0032	0.9974 ± 0.0004	0.0237 ± 0.0048	0.0522 ± 0.0040
LMH (PoE)	0.9435 ± 0.0093	0.9596 ± 0.0070	0.9849 ± 0.0113	0.0621 ± 0.0160	0.0685 ± 0.0213
Noise-Consistency	0.9346 ± 0.0075	0.9560 ± 0.0039	0.9960 ± 0.0013	0.0347 ± 0.0137	0.0655 ± 0.0130
GACR-B (exploratory)	0.9417 ± 0.0059	0.9610 ± 0.0042	0.9972 ± 0.0005	0.0245 ± 0.0023	0.0540 ± 0.0038
Adaptive-CADQ	0.9388 ± 0.0072	0.9610 ± 0.0044	0.9972 ± 0.0008	0.0241 ± 0.0070	0.0527 ± 0.0028

Baseline per-class F1 was 0.9755 ± 0.0032 for Normal, 0.9624 ± 0.0055 for Pneumonia, and 0.8810 ± 0.0174 for Pneumothorax. LMH increased Pneumothorax F1 to 0.9003 ± 0.0191 but reduced Pneumonia F1 to 0.9575 ± 0.0077 . Noise-Consistency had the widest Pneumothorax variation (0.8753 ± 0.0237). Adaptive-CADQ showed the most stable Pneumonia F1 (0.9616 ± 0.0029). The distributions and the retained control contrast are shown in Figure 2.

No condition improved Macro-F1 over baseline after Benjamini-Hochberg correction (Table 2). The largest unadjusted Macro-F1 signal was the exploratory GACR-B contrast ($\Delta = 0.0020$, Wilcoxon $p = 0.0547$; adjusted $p = 0.2188$). LMH and Noise-Consistency had lower seed-42 Pneumothorax AUROC than baseline; their significant DeLong results indicate degradation, not improvement. Adaptive-CADQ showed no AUROC difference.

Table 2

Paired comparisons with baseline. Wilcoxon uses eight paired seed-level Macro-F1 values. DeLong and McNemar use the retained seed-42 probability and prediction arrays for the same 455 studies. Values in parentheses are Benjamini-Hochberg adjusted within each test family. McNemar n_{01}/n_{10} are the discordant counts; $p = 1$ does not imply identical predictions.

Condition	Wilcoxon Macro-F1 p (BH)	DeLong PNX-AUROC p (BH)	McNemar n_{01}/n_{10} ; p (BH)
LMH (PoE)	0.2500 (0.3333)	2.9×10^{-6} (1.2×10^{-5})	27/31; 0.6936 (1.0000)
Noise-Consistency	0.1953 (0.3333)	0.0022 (0.0044)	32/49; 0.0754 (0.3018)
GACR-B (exploratory)	0.0547 (0.2188)	0.0315 (0.0420)	9/8; 1.0000 (1.0000)
Adaptive-CADQ	0.8438 (0.8438)	0.7304 (0.7304)	15/14; 1.0000 (1.0000)

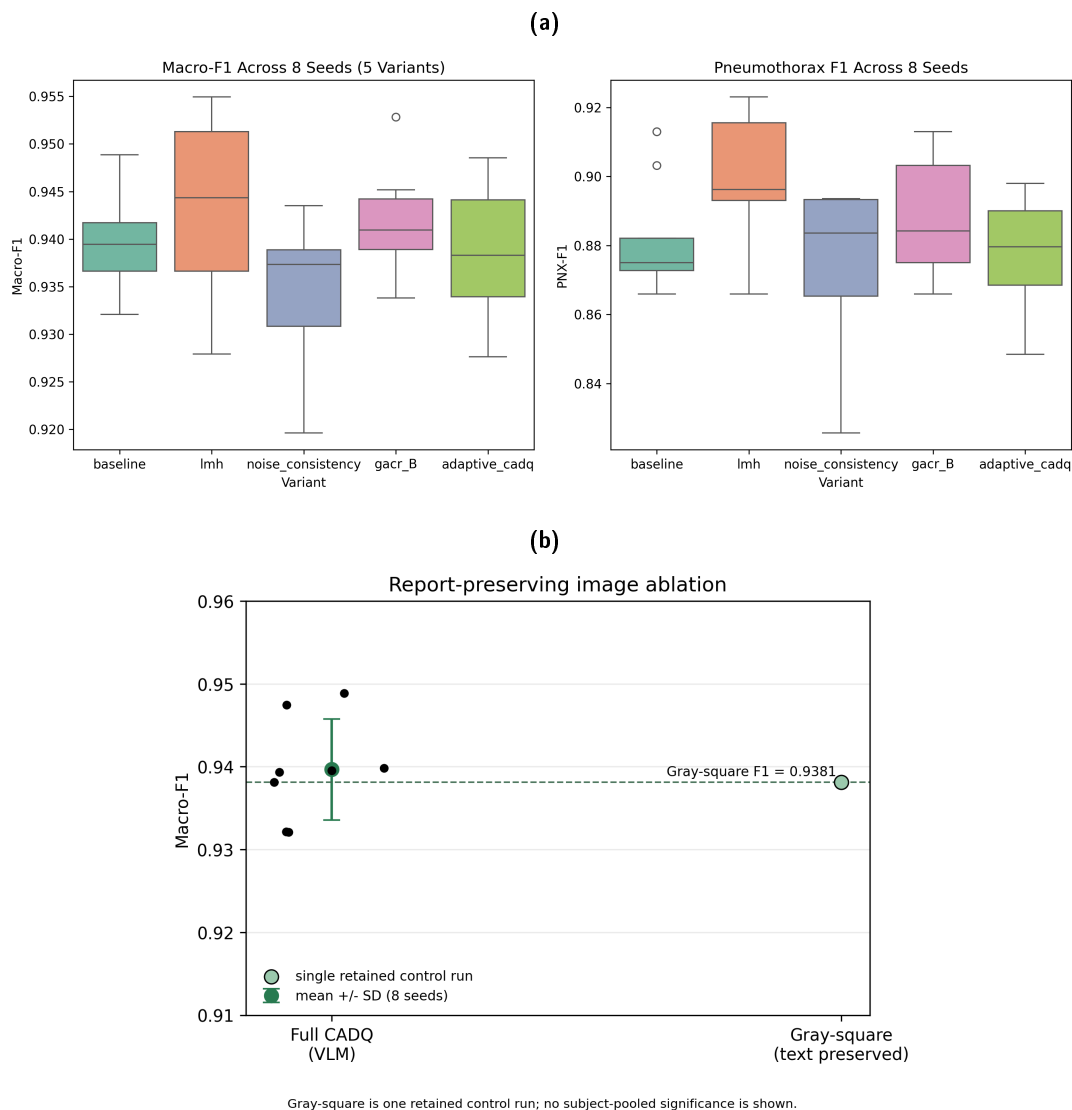


Figure 2: Predictive performance and controls. (a) Macro-F1 and Pneumothorax-F1 distributions over eight seeds for the five fusion conditions. (b) Retained single-run controls compared with the baseline VLM. Image-only checkpoint selection was imperfect, and control values are descriptive rather than modality rankings.

The balanced discordance for GACR-B and Adaptive-CADQ is informative but less extreme than the stale interpretation in the original draft. GACR-B disagreed with baseline on 17 studies (9 versus 8 across the two off-diagonal directions), and Adaptive-CADQ on 29 (15 versus 14). McNemar $p = 1.000$ indicates no evidence of directional asymmetry in these paired errors; it does not indicate byte-identical predictions. The confusion matrices and receiver-operating-characteristic curves in Figure 3 are descriptive views of the same repeated test cohort.

Calibration separated some conditions that had similar Macro-F1. Baseline ECE was 0.0237 ± 0.0048 and Brier score 0.0522 ± 0.0040 . LMH increased ECE to 0.0621 ± 0.0160 (2.6-fold) and Brier score to 0.0685 ± 0.0213 ; the adjusted seed-level p -values were 0.0313 and 0.0156. Noise-Consistency increased Brier score to 0.0655 ± 0.0130 (adjusted $p = 0.0156$), whereas its ECE change was inconclusive. GACR-B and Adaptive-CADQ remained close to baseline. Figure 4 shows the baseline reliability diagram and seed-level ECE distributions.

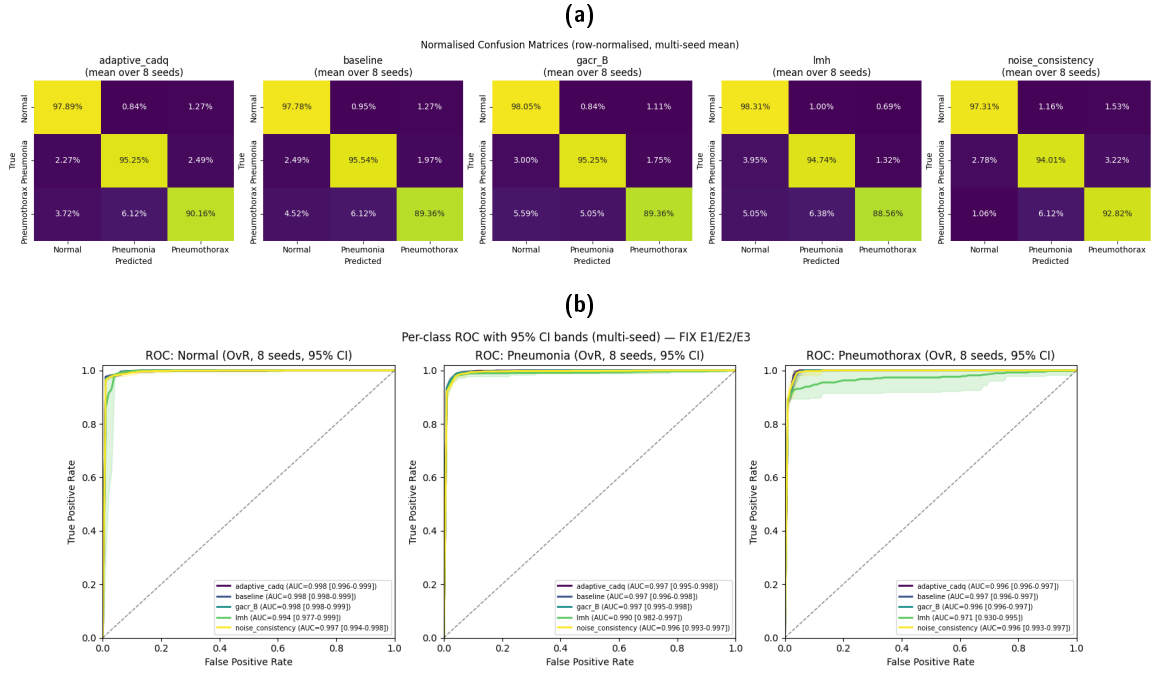


Figure 3: Class-specific error and discrimination. (a) Row-normalized confusion matrices aggregated descriptively over seeds; every row is normalized by its true class. (b) Per-class receiver-operating-characteristic curves with descriptive multi-seed 95% bands. Significant DeLong results for LMH, Noise-Consistency, and GACR-B reflect lower seed-42 Pneumothorax AUROC than baseline, not improvement. DeLong inference was not performed on subjects pooled across seeds.

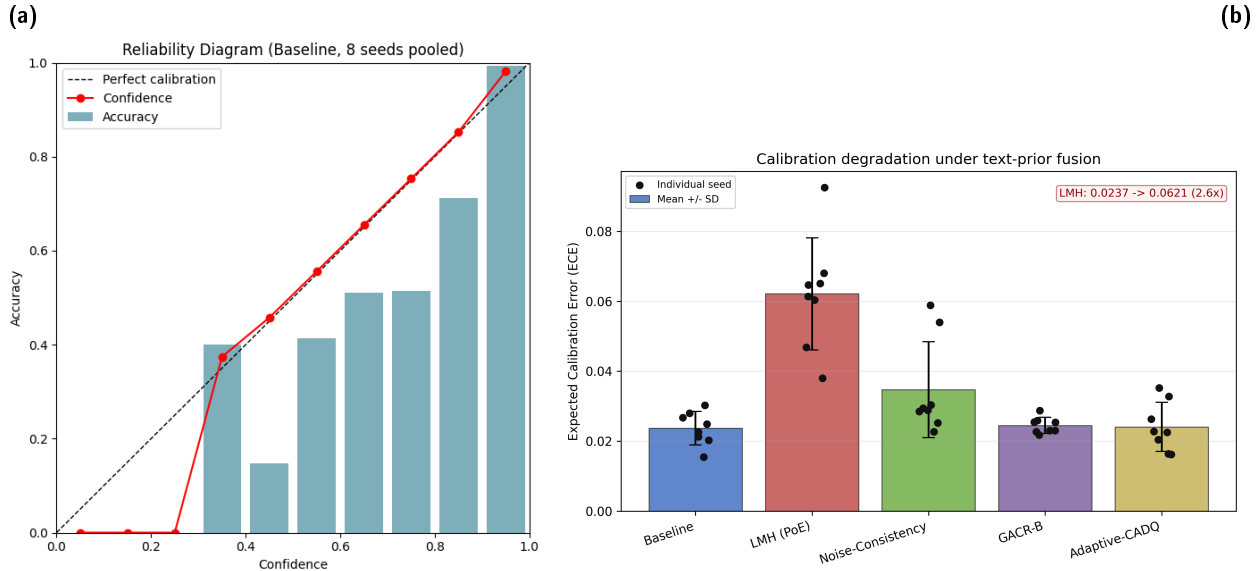


Figure 4: Calibration audit. (a) Descriptive baseline reliability diagram. (b) ECE by condition with individual seed values and mean $\pm$ standard deviation. LMH increased ECE from 0.0237 to 0.0621.

4.2. Report-preserving controls and ICM

The retained image-only control achieved Macro-F1 0.4611, accuracy 0.5670, Macro-AUROC 0.6728, ECE 0.0993, and Brier score 0.6142. The text-only control achieved 0.8555, 0.9077, 0.9820, 0.0203, and 0.1289, respectively. The Gray-Square control achieved Macro-F1 0.9381, accuracy 0.9604, Macro-AUROC 0.9960, ECE 0.0369, and Brier score 0.0589. Its Macro-F1 was only 0.0016 below the baseline eight-seed mean. Because these were single

retained runs and the image-only checkpoint was not selected consistently, they support an audit of the archived system but not a definitive ranking of modality capacity.

Baseline ICM was 0.0016 ± 0.0065 with median 0.0014 and bootstrap 95% confidence interval $[-0.0026, 0.0061]$ (Table 3). The signed-rank test did not reject zero ($p = 0.4219$), while TOST supported equivalence inside the operational interval ($p = 0.0040$). LMH had ICM 0.0056 ± 0.0098 and did not meet the equivalence criterion. Noise-Consistency had a negative mean ICM (-0.0039 ± 0.0081), whereas exploratory GACR-B and Adaptive-CADQ had small positive values. None showed a robust positive contribution relative to the Gray-Square reference.

Table 3

Image Contribution Metric statistics over eight seeds. Confidence intervals are percentile bootstrap intervals from 2,000 resamples. Wilcoxon tests compare ICM with zero; TOST evaluates equivalence within the operational interval $[-0.01, 0.01]$.

Condition	Mean	SD	Median	95% CI	Wilcoxon p	TOST p	Cohen's d
Baseline	0.0016	0.0065	0.0014	$[-0.0026, 0.0061]$	0.4219	0.0040	0.247
LMH (PoE)	0.0056	0.0098	0.0066	$[-0.0011, 0.0118]$	0.1953	0.1223	0.566
Noise-Consistency	-0.0039	0.0081	-0.0009	$[-0.0097, 0.0009]$	0.3828	0.0345	-0.476
GACR-B (exploratory)	0.0038	0.0062	0.0030	$[-0.0002, 0.0081]$	0.1953	0.0122	0.607
Adaptive-CADQ	0.0007	0.0077	0.0002	$[-0.0043, 0.0057]$	0.8438	0.0054	0.089

The seed pairing, equivalence interval, and retained gate values are shown in Figure 5. The Gray-Square value was reused for all eight ICM calculations; its run-to-run uncertainty is therefore absent from the displayed intervals. The equivalence result should be interpreted as a property of the retained comparator, not an estimate from two independently replicated arms.

4.3. Gate behaviour and qualitative inspection

The retained baseline gate means were 0.42624 for Normal, 0.49963 for Pneumonia, and 0.57336 for Pneumothorax. The overall mean was 0.49975 compared with an initialization mean of 0.50000, and the largest per-seed class-wise deviation from the equation-derived initialization was 0.000499. The gate did not collapse toward its lower bound of 0.20, nor did the retained values show a meaningful shift toward greater image reliance. We call this descriptive pattern *gate stagnation*; it is evidence that this parameter changed little, not proof that its gradient was exactly zero or that the image branch was disconnected. Per-seed values appear in Supplementary Table S1.

The three retained Grad-CAM examples showed diffuse activation in Baseline seed 42 (Supplementary Fig. S1). They were not compared with localization annotations, deletion curves, or clinician assessment. They therefore provide only a qualitative sanity check and cannot establish visual grounding or its absence.

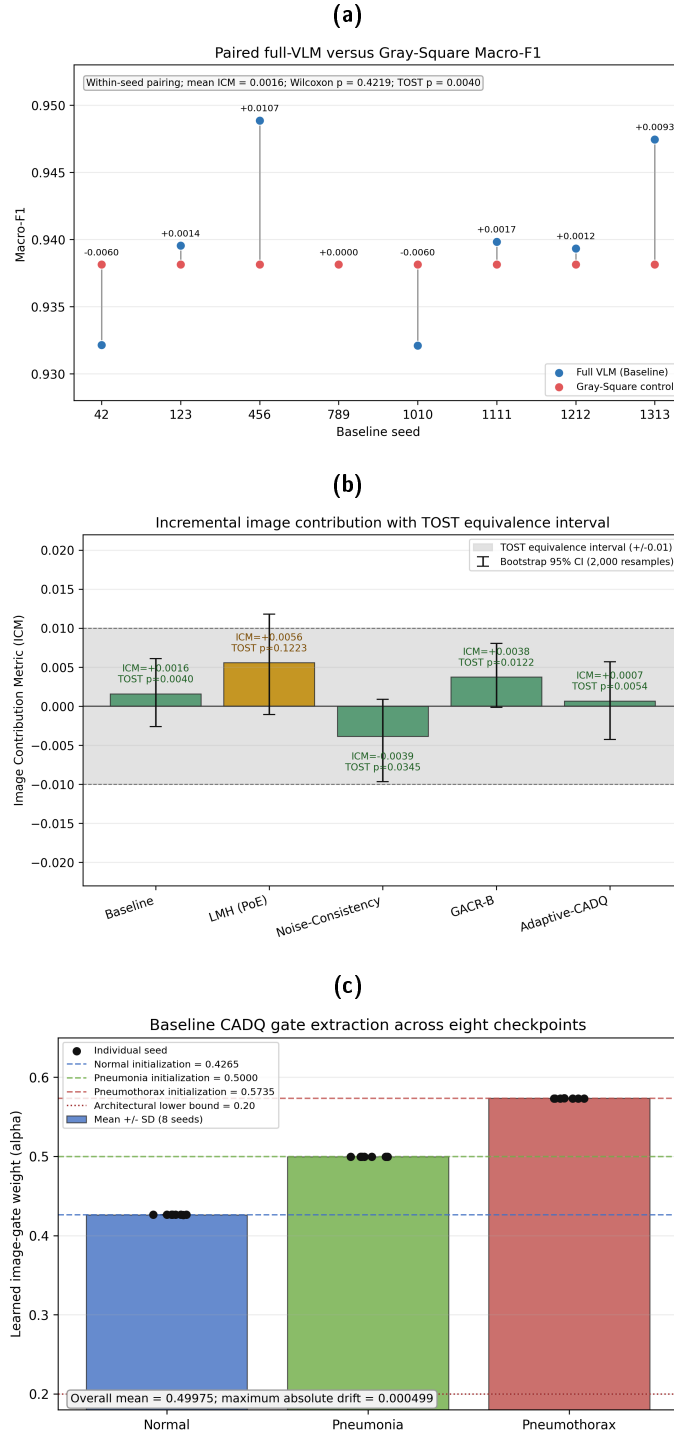


Figure 5: Modality-contribution and gate audit. (a) Baseline VLM Macro-F1 at each seed paired with the retained Gray-Square reference; lines show the small differences used to construct ICM. (b) Mean ICM with percentile bootstrap 95% confidence intervals; shading marks the operational TOST interval $[-0.01, 0.01]$. (c) Baseline class-specific image gates from eight retained checkpoint exports. Dots are individual seeds, bars show mean $\pm$ standard deviation, and dashed lines are the equation-derived initial values.

5. Discussion

5.1. Principal findings and interpretation

The central result is a mismatch between high predictive performance and incremental image evidence. Baseline Macro-F1 approached 0.94, yet a constant gray image paired with the same report retained Macro-F1 0.9381. Baseline ICM was close to zero, its signed-rank test was non-significant, and TOST supported equivalence within the project-defined interval. None of the four alternative fusion or text-debiasing conditions produced a robust ICM increase. The LMH text-prior intervention had the highest mean Macro-F1 but lower AUROC and much worse calibration. The retained baseline gates stayed close to initialization. Together, these findings show that aggregate performance, a non-collapsed gate, and visually plausible activation maps can coexist with negligible performance change after image replacement.

The most direct explanation is construct overlap. MIMIC-CXR pairs each image with a report, CheXpert uses report language to generate the target, and the model receives the report’s findings field [1, 11]. The Gray-Square control preserves this high-information route. It does not show that the CheXpert labels are clinically wrong, that every prediction copied a particular phrase, or that radiographs lack diagnostic value. It shows that the evaluated endpoint does not isolate independent image-based recognition. The claim supported by the data is therefore narrow: under this report-linked setup and archived implementation, the measured incremental image contribution was negligible.

We use *modality inertia* to describe the combined observation that the output changes little under report-preserving image replacement while the static gate remains near its starting value. This is a useful diagnostic label, but the current experiment does not identify a unique mechanism. Redundant features, optimization dynamics, precision, normalization mismatch, or weak image gradients could each contribute. The gate result should not be promoted from a parameter-stability observation to proof of causal gradient starvation without gradient traces or controlled retraining.

The debiasing results clarify why endpoint design matters. Product-of-experts methods are intended to suppress an easy language prior. Here, however, report language helped generate the target. Down-weighting this pathway can degrade AUROC or calibration without creating new image information, as observed for LMH. Noise-Consistency cannot support its intended interpretation because the archived negative-KL sign rewards divergence. Adaptive-CADQ changed the gate form but not the endpoint. Exploratory GACR-B, following the grounding-aware proposal of Sadanandan and Behzadan [9], achieved a small ICM but not a robust Macro-F1 improvement. The experiment did not measure whether its $\lambda = 0.3$ penalty generated insufficient gradient; that mechanism requires a dedicated ablation and is not inferred from the null performance result.

The corrected McNemar interpretation is also important. GACR-B and Adaptive-CADQ had $p = 1.000$, but both had non-zero discordant predictions. The off-diagonal counts were nearly balanced, so McNemar detected no directional change in error rates. This supports a conclusion of similar aggregate paired behaviour, not byte-identical decision boundaries. Reporting the contingency counts prevents a non-significant symmetry test from being mistaken for an identity test.

5.2. Implications for biomedical informatics

Multimodal evaluation should make modality contribution explicit. A minimum audit should document how targets were produced, whether any input shares that source, what happens when each modality is replaced or permuted, whether controls are matched across seeds, and how calibration changes. The ablation must preserve all other inputs and the evaluation cohort. Where the scientific claim is that an image contributes, an independent or image-derived reference standard is preferable to a target extracted from the accompanying report. Counterfactual text tests, temporally separated note fields, and cross-patient report swaps can further reveal which source carries the prediction.

ICM is one compact summary of such an intervention. Its advantages are an interpretable direction and a model-relative scale, but it inherits every limitation of the chosen performance metric and comparator. Macro-F1 may hide class-specific harm; a single Gray-Square run understates control uncertainty; and the $[-0.01, 0.01]$ interval has not been clinically validated. ICM should therefore be reported with absolute performance, per-class outcomes, calibration, confidence intervals, and a description of the control. Validation on independent-label tasks is needed before comparing ICM across datasets.

Gate and saliency analyses should remain secondary. A gate is shaped by parameterization and can be unidentifiable when modality signals are redundant. The one-key attention block used here cannot align words to image regions, and three Grad-CAM examples lack quantitative localization evidence. A stronger future design would use token- and

region-level interactions, paired modality interventions, gradient and gate trajectories, concept localization, and reader-validated explanations. Even then, behavioural change under input intervention remains the more direct evidence of contribution.

For regulatory-facing evidence, the central lesson is traceability rather than a proposed mandatory threshold. Good machine-learning-practice and real-world- evidence principles emphasize representative evaluation, transparent data provenance, uncertainty, and evidence aligned with intended use [34, 35]. A multimodal device dossier should therefore state whether labels and inputs share provenance and include matched ablations when modality use is part of the intended claim. ICM and gate inspection can support that evidence package, but neither substitutes for external validation, clinical utility, subgroup performance, or human-factors assessment.

5.3. Limitations and future work

This audit has substantial limitations. It used 2,953 studies from one database, one labeler, three classes, and only 47 Pneumothorax test studies. The uncertain-to-negative mapping, class caps, prevalence, and inherited split may affect the results. No external cohort, radiologist-adjudicated reference, prospective workflow, reader study, subgroup analysis, decision-curve analysis, or outcome evaluation was available. Generalization to other reports, institutions, devices, and label processes is unknown.

Implementation defects limit causal interpretation. Image values were mapped to $[-1, 1]$, which may not match the native TorchXRayVision convention for the selected weights. The Noise-Consistency objective had the wrong KL sign. The attention block used one token per modality. Image-only checkpoint restoration was inconsistent. These defects cannot be repaired by manuscript wording; the affected experiments require rerunning. The current findings audit the archived system and should not be presented as a definitive test of correctly implemented noise consistency or optimized image-only performance.

Control and statistical asymmetries also matter. The same single Gray-Square run was reused to construct eight ICM values, so comparator uncertainty was not propagated. Eight seeds provide a small replication sample. DeLong and McNemar results were computed only from the retained seed-42 arrays, whereas seed-aggregated curves are descriptive. Adaptive-CADQ's eight-seed extension and GACR-B were post hoc. No a priori sample-size calculation established power for a clinically meaningful ICM. Future studies should train every control at every seed, preregister the equivalence margin and estimand, and use a larger independent-label cohort.

Finally, the result archive was assembled from multiple notebook outputs. The merge retained first-seen duplicates without a documented source-priority rule or content hashes, and some probabilities had to be recovered from a companion pickle. A confirmatory release should include a locked environment, dataset and split hashes, notebook and checkpoint hashes, deterministic merge rules, full probability arrays, and a machine-readable provenance manifest. The present manifest is descriptive and identifies rather than resolves these issues.

6. Conclusion

In this MIMIC-CXR-derived task, replacing the radiograph with a constant image preserved nearly all baseline Macro-F1. Baseline ICM was near zero and equivalent within the operational interval, the tested interventions did not demonstrate a robust increase in incremental image contribution, and retained static gates remained near initialization. These findings expose a benchmark-validity risk created when labels and text inputs share report provenance. They do not show that radiographs or medical VLMs are generally uninformative. Independent labels, corrected implementations, matched multi-seed controls, provenance locking, and external validation are required. ICM is best used as one component of a broader modality-reliance audit.

Ethics approval and consent to participate

MIMIC-CXR is de-identified and was accessed through official PhysioNet credentialing under its Data Use Agreement. No participants were recruited, no direct identifiers were accessed, and consent was not applicable. The source resource was released under the originating institutional review-board approvals; this secondary computational analysis introduced no intervention or participant contact.

Data availability

Raw MIMIC-CXR images and reports remain under PhysioNet access controls and are not redistributed. Non-identifying summaries, code, manuscript source, and figure-generation materials are available at <https://github.com/i-mTanvir/MediVision>. Cohort identifiers can be shared only in a manner consistent with the PhysioNet Data Use Agreement.

Code availability

The analysis notebooks, LaTeX source, figure scripts, and environment records are available at <https://github.com/i-mTanvir/MediVision>. A permanent archival DOI will be added if a versioned release is deposited. Credentials, raw reports, and restricted identifiers are excluded.

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Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have influenced this work.

Author contributions

Nafis Al Rahman and Tanvir Mahmud contributed equally to conceptualization, methodology, investigation, software, data curation, formal analysis, validation, visualization, and writing. Both authors reviewed and approved the manuscript and accept responsibility for its content.

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Declaration of generative AI and AI-assisted technologies in the writing process

Generative-AI tools were not used to generate study data, execute experiments, create scientific results, or make analytical decisions. Any language or formatting assistance used during manuscript preparation was reviewed and edited by both authors, who take full responsibility for the final content.

S. Supplementary material

S.1. Per-seed gate values

Table S1

Baseline CADQ image-gate values retained in the checkpoint export. Initialization values are (0.426524, 0.500000, 0.573476). Maximum drift is the largest absolute class-wise difference from initialization for that seed.

Seed	Normal α	Pneumonia α	Pneumothorax α	Max drift
42	0.426270	0.499512	0.573242	0.000488
123	0.426270	0.499512	0.573730	0.000488
456	0.426025	0.499756	0.573242	0.000499
789	0.426270	0.499756	0.573242	0.000255
1010	0.426270	0.499756	0.573242	0.000255
1111	0.426270	0.499512	0.573242	0.000488
1212	0.426270	0.499756	0.573730	0.000255
1313	0.426270	0.499512	0.573242	0.000488

S.2. Training configuration

Table S2

Training hyperparameters for the archived experiments.

Setting	Value
Image encoder	TorchXRayVision DenseNet-121, res224 MIMIC weights
Text encoder	Bio_ClinicalBERT; first four transformer layers frozen
Optimizer	AdamW; weight decay 10^{-4}
Learning rates	10^{-5} image/text; 10^{-4} heads; 5×10^{-5} static gates
Schedule	OneCycleLR with 10% warm-up; 10 epochs
Batching	Physical batch 8; four accumulation steps; effective batch 32
Objective	Focal cross-entropy ($\gamma = 1.5$), label smoothing 0.05, effective-number $\beta = 0.99$
Precision	FP16 mixed precision with gradient scaling
Gate initialization	$u = (-0.5, 0, 0.5)$; $\alpha = (0.426524, 0.500000, 0.573476)$
Augmentation	Crop, flip, rotation, translation, intensity jitter; minority-class transform
Seeds	42, 123, 456, 789, 1010, 1111, 1212, 1313
Hardware	Kaggle NVIDIA T4 two-GPU environment

S.3. Reporting-checklist summaries

Table S3

CLAIM 2024 audit summary. Counts reproduce the archived 27-item project audit; full item-level responses remain in the repository.

Status	Count	Summary
Yes	13	Study type, source, outcome, model description, metrics, comparisons, limitations, and seed-level analysis were reported.
Partial	10	External validation, comparator breadth, ablation coverage, and reproducibility were limited by scope and controlled data access.
No	4	Cross-validation, external state-of-the-art baselines, pretraining-overlap audit, and complete credentialing documentation were not included.

Table S4

TRIPOD+AI audit summary. Counts reproduce the archived 41-item project audit; full item-level responses remain in the repository.

Status	Count	Summary
Yes	9	The title, data source, participants, model, statistical tests, limitations, registration history, and availability statements were addressed.
Partial	18	Abstract detail, hyperparameters, comparison breadth, reproducibility, and ethics reporting were only partly addressed.
No	14	External validation, decision-curve analysis, subgroup/fairness analysis, missing-data assessment, calibration transfer, and a model card were absent.

S.4. Provenance manifest

Table S5

Condition-level provenance and known constraints in the fixed archive.

Condition	Seed coverage	Retained outputs	Known constraint
Baseline CADQ	8 seeds	Metrics, predictions, probabilities, gates	Image normalization requires confirmation against XRV convention
LMH (PoE)	8 seeds	Metrics, predictions, probabilities	Warm-started; calibration degraded
Noise-Consistency	8 seeds	Metrics, predictions, probabilities	Negative KL sign rewards divergence
GACR-B	8 seeds	Metrics, predictions, recovered probabilities	Added post hoc; merge provenance was not hash-locked
Adaptive-CADQ	8 seeds	Metrics, predictions, probabilities	Extended from three to eight seeds post hoc
Controls	One retained run each	Metrics and selected arrays	Gray-Square uncertainty not replicated; image-only best-checkpoint restoration inconsistent

S.5. Seed-42 sensitivity intervals

Table S6

Seed-42 per-class sensitivity with Wilson 95% confidence intervals ($n = 455$; Normal $n = 240$, Pneumonia $n = 168$, Pneumothorax $n = 47$).

Condition	Normal	Pneumonia	Pneumothorax
Baseline	0.975 [0.946, 0.988]	0.947 [0.903, 0.972]	0.894 [0.774, 0.954]
LMH (PoE)	0.962 [0.929, 0.980]	0.947 [0.903, 0.972]	0.894 [0.774, 0.954]
Noise-Consistency	0.962 [0.929, 0.980]	0.947 [0.903, 0.972]	0.894 [0.774, 0.954]
GACR-B	0.975 [0.946, 0.988]	0.947 [0.903, 0.972]	0.894 [0.774, 0.954]
Adaptive-CADQ	0.979 [0.952, 0.991]	0.953 [0.910, 0.976]	0.894 [0.774, 0.954]

S.6. Qualitative saliency examples

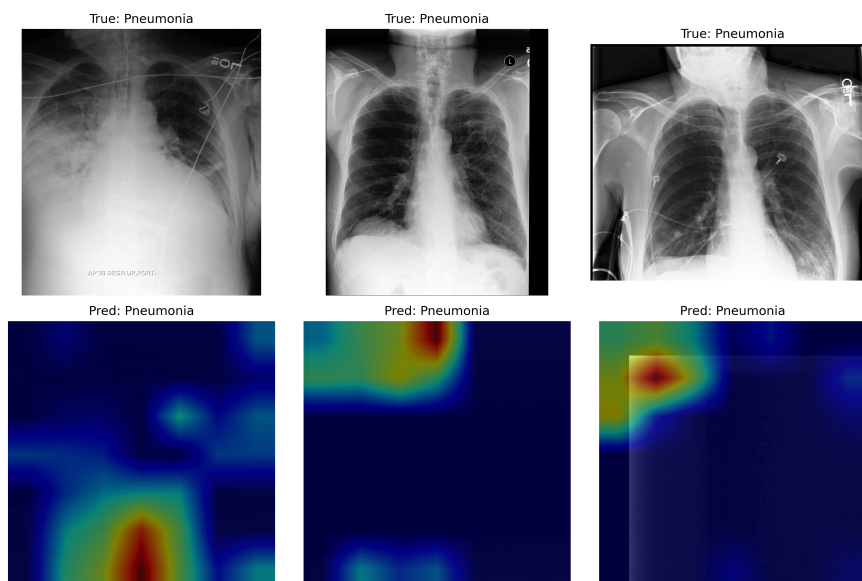


Figure S1: Qualitative Grad-CAM examples for Baseline seed 42. The examples were not evaluated against localization annotations and do not support a causal or quantitative grounding claim.

References

- [1] A. E. W. Johnson, T. J. Pollard, S. J. Berkowitz, N. R. Greenbaum, M. P. Lungren, C.-Y. Deng, R. G. Mark, S. Horng, Mimic-cxr, a de-identified publicly available database of chest radiographs with free-text reports, *Scientific Data* 6 (2019) 317. doi:10.1038/s41597-019-0322-0.
- [2] Y. Zhang, H. Jiang, Y. Miura, C. D. Manning, C. P. Langlotz, Contrastive learning of medical visual representations from paired images and text, *Proceedings of Machine Learning Research* 182 (2022).
- [3] S.-C. Huang, S. Shen, et al., Gloria: A multimodal global-local representation learning framework for label-efficient medical image recognition, *Proceedings of the IEEE/CVF International Conference on Computer Vision* (2021).

- [4] B. Boecking, N. Usuyama, S. Bannur, D. C. Castro, A. Schwaighofer, S. Hyland, M. Wetscherek, T. Naumann, A. Nori, J. Alvarez-Valle, H. Poon, O. Oktay, Making the most of text semantics to improve biomedical vision-language processing, arXiv preprint arXiv:2204.09817 (2022).
- [5] R. Geirhos, et al., Shortcut learning in deep neural networks, *Nature Machine Intelligence* (2020).
- [6] I. Banerjee, K. Bhattacharjee, J. L. Burns, H. Trivedi, S. Purkayastha, L. Seyyed-Kalantari, B. N. Patel, S. Rakesh, J. Gichoya, Shortcuts causing bias in radiology artificial intelligence: Causes, evaluation, and mitigation, *Journal of the American College of Radiology* (2023).
- [7] Q. Jin, et al., Hidden flaws behind expert-level accuracy of multimodal gpt-4 vision in medicine, *npj Digital Medicine* (2024).
- [8] M. Lofthia, S. Ziegelmayer, L. Adams, D. Truhn, A. Maier, S. Tayebi Arasteh, Vision-language models for chest radiography do not always need the image, arXiv preprint arXiv:2606.17710 (2026).
- [9] B. Sadanandan, V. Behzadan, Consistent but dangerous: Per-sample safety classification reveals false reliability in medical vision-language models, arXiv preprint arXiv:2603.20985 (2026).
- [10] A. Wang, J. Zhang, M. Oberst, Revisiting performance claims for chest x-ray models using clinical context, *Proceedings of Machine Learning Research* 333 (2026).
- [11] J. Irvin, P. Rajpurkar, M. Ko, Y. Yu, S. Ciurea-Ilcus, C. Chute, H. Marklund, B. Haghighi, R. L. Ball, K. S. Shpanskaya, J. Seekins, D. A. Mong, S. S. Halabi, J. K. Sandberg, R. Jones, D. B. Larson, C. P. Langlotz, B. N. Patel, M. P. Lungren, A. Y. Ng, Chexpert: A large chest radiograph dataset with uncertainty labels and expert comparison, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 33, 2019, pp. 590–597. doi:10.1609/AAAI.V33I01.3301590.
- [12] C. Sabottke, J. Lee, A. Chiang, B. Spieler, R. Mushtaq, Text report analysis to identify opportunities for optimizing target selection for chest radiograph artificial intelligence models, *Journal of Imaging Informatics in Medicine* 37 (2024) 402–411.
- [13] S. M. Santomartino, J. R. Zech, K. Hall, J. Jeudy, V. Parekh, P. H. Yi, Evaluating the performance and bias of natural language processing tools in labeling chest radiograph reports, *Radiology* (2024).
- [14] A. Agrawal, D. Batra, D. Parikh, A. Kembhavi, Don't just assume; look and answer: Overcoming priors for visual question answering, *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (2018).
- [15] C. Clark, M. Yatskar, L. Zettlemoyer, Don't take the easy way out: Ensemble based methods for avoiding known dataset biases, arXiv preprint arXiv:1909.03683 (2019).
- [16] R. Cadene, C. Dancette, M. Cord, D. Parikh, et al., Rubi: Reducing unimodal biases for visual question answering, arXiv preprint arXiv:1906.10169 (2019).
- [17] L.-F. Chiang, An interpretable chest x-ray classification framework using prototype memory and counterfactual consistency, *Cureus* 18 (2) (2026) e103134.
- [18] J. Arevalo, T. Solorio, M. Montes-y Gómez, F. A. González, Gated multimodal units for information fusion, arXiv preprint arXiv:1702.01992 (2017).
URL link
- [19] D. Lakens, Equivalence tests: A practical primer for t tests, correlations, and meta-analyses, *Social Psychological and Personality Science* 8 (4) (2017) 355–362. doi:10.1177/1948550617697177.
- [20] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra, Grad-cam: Visual explanations from deep networks via gradient-based localization, in: *2017 IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 618–626. doi:10.1109/ICCV.2017.74.
- [21] E. Alsentzer, J. Murphy, W. Boag, W.-H. Weng, D. Jindl, T. Naumann, M. McDermott, Publicly available clinical bert embeddings, in: *Proceedings of the 2nd Clinical Natural Language Processing Workshop, Association for Computational Linguistics*, 2019, pp. 72–78. doi:10.18653/v1/W19-1909.
- [22] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, Bert: Pre-training of deep bidirectional transformers for language understanding, in: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Vol. 1, Association for Computational Linguistics*, 2019, pp. 4171–4186. doi:10.18653/v1/N19-1423.
- [23] A. Myronenko, D. Yang, B. Turkbey, M. Aboian, S. Azamat, E. Akcicek, H. Yin, P. Molchanov, M. Edgar, Y. He, P. Guo, Y. Tang, D. Xu, Reasoning visual language model for chest x-ray analysis, arXiv preprint arXiv:2510.23968 (2025).
- [24] J. P. Cohen, J. D. Viviano, P. Bertin, P. Morrison, P. Torabian, M. Guarrera, M. P. Lungren, A. Chaudhari, R. Brooks, M. Hashir, H. Bertrand, Torchxrayvision: A library of chest x-ray datasets and models, in: *Proceedings of the 5th International Conference on Medical Imaging with Deep Learning, Vol. 172 of Proceedings of Machine Learning Research, PMLR*, 2022, pp. 231–249.
URL link
- [25] G. Huang, Z. Liu, L. van der Maaten, K. Q. Weinberger, Densely connected convolutional networks, in: *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 2261–2269. doi:10.1109/CVPR.2017.243.
- [26] I. Loshchilov, F. Hutter, Decoupled weight decay regularization, in: *7th International Conference on Learning Representations*, 2019.
URL link
- [27] L. N. Smith, N. Topin, Super-convergence: Very fast training of neural networks using large learning rates, *Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications* 11006 (2019) 1100612. doi:10.1117/12.2520589.
- [28] T.-Y. Lin, P. Goyal, R. Girshick, K. He, P. Dollár, Focal loss for dense object detection, in: *2017 IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 2999–3007. doi:10.1109/ICCV.2017.324.
- [29] E. R. DeLong, D. M. DeLong, D. L. Clarke-Pearson, Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach, *Biometrics* 44 (3) (1988) 837–845. doi:10.2307/2531595.
- [30] Q. McNemar, Note on the sampling error of the difference between correlated proportions or percentages, *Psychometrika* 12 (2) (1947) 153–157. doi:10.1007/BF02295996.
- [31] Y. Benjamini, Y. Hochberg, Controlling the false discovery rate: A practical and powerful approach to multiple testing, *Journal of the Royal Statistical Society: Series B (Methodological)* 57 (1) (1995) 289–300. doi:10.1111/j.2517-6161.1995.tb02031.x.
- [32] J. Mongan, L. Moy, B. J. Erickson, et al., Checklist for artificial intelligence in medical imaging (claim): A guide for authors and reviewers, *Radiology: Artificial Intelligence* 6 (2) (2024) e230120. doi:10.1148/ryai.230120.
- [33] G. S. Collins, K. G. M. Moons, P. Dhiman, et al., Tripod+ai statement: Updated guidance for reporting prediction models that use regression or machine learning methods, *The BMJ* 385 (2024) e078378. doi:10.1136/bmj-2023-078378.
- [34] U.S. Food and Drug Administration, Good machine learning practice for medical device development: Guiding principles, FDA guidance, accessed 2026-08-31 (2025).
URL link
- [35] U.S. Food and Drug Administration, Use of real-world evidence to support regulatory decision-making for medical devices, FDA guidance, accessed 2026-08-31 (2025).
URL link